

Project Documentation

Heart Disease Analysis Dashboard using Tableau

Step 1: Project Selection

Description

The project selected for this internship is Heart Disease Analysis using Tableau. The main objective is to analyse healthcare data and identify the factors contributing to heart disease through interactive visualizations.

Output:

- Project topic finalized.
- Objectives identified.

Step 2: Dataset Collection

Description

The Heart Disease 2020 dataset was collected. The dataset contains patient demographic details, lifestyle information, and medical conditions.

Dataset Fields

- HeartDisease
- BMI
- Smoking
- Alcohol Drinking
- Stroke
- Physical Health
- Mental Health
- Sex
- Age Category

- Race
- Diabetic
- Physical Activity
- General Health
- Sleep Time
- Asthma
- Kidney Disease
- Skin Cancer

Output:

Dataset ready for analysis.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	HeartDisease	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	No	20.34	No	No	Yes	0	0	No	Female	55-59	white	Yes	Yes	Very good	5	Yes	No	No	No	No
3	No	20.34	No	No	Yes	0	0	No	Female	60 or older	white	No	Yes	Very good	7	No	No	No	No	No
4	No	20.34	Yes	No	No	20	20	No	Male	45-49	white	Yes	Yes	Fair	0	Yes	No	No	No	No
5	No	24.21	No	No	No	0	0	No	Female	75-79	white	No	No	Good	6	No	No	Yes	No	No
6	No	23.71	No	No	No	28	0	Yes	Female	40-44	white	No	Yes	Very good	8	No	No	No	No	No
7	Yes	28.87	Yes	No	No	0	0	Yes	Female	75-79	Black	No	No	Fair	22	No	No	No	No	No
8	Yes	21.68	No	No	No	15	0	No	Female	70-74	white	No	Yes	Fair	4	Yes	No	Yes	No	No
9	No	21.94	Yes	No	No	5	0	Yes	Female	60 or older	white	Yes	No	Good	9	Yes	No	No	No	No
10	No	25.40	No	No	No	0	0	No	Female	60 or older	white	No, border	No	Fair	5	No	Yes	No	No	No
11	No	40.87	No	No	No	0	0	Yes	Male	65-69	white	No	Yes	Good	20	No	No	No	No	No
12	Yes	34.3	Yes	No	No	30	0	Yes	Male	60-64	white	Yes	No	Poor	15	Yes	No	No	No	No
13	No	26.71	Yes	No	No	0	0	No	Female	55-59	white	No	Yes	Very good	5	No	No	No	No	No
14	No	24.17	Yes	No	No	0	0	Yes	Male	75-79	white	Yes	Yes	Very good	8	No	No	No	No	No
15	No	25.15	No	No	No	7	0	Yes	Female	60 or older	white	No	No	Good	7	No	No	No	No	No
16	No	25.28	Yes	No	No	0	10	Yes	Female	60-64	white	No	No	Good	5	No	No	No	No	No
17	No	28.18	No	No	No	1	0	No	Female	50-54	white	No	Yes	Very good	6	No	No	No	No	No
18	No	20.20	No	No	No	5	2	No	Female	70-74	white	No	No	Very good	20	No	No	No	No	No

Step 3: Import Dataset into Tableau

Procedure

1. Open Tableau Desktop.
2. Click Connect.
3. Select Text File.
4. Choose the Heart Disease dataset.
5. Verify all fields.
6. Click Sheet 1.

Output

Dataset successfully imported into Tableau.

Name	Age	Sex	BMI	Smoking	Diabetes	Physical Health	Mental Health	Heart Disease
John Doe	45	Male	25.5	Yes	No	Good	Good	No
Jane Smith	38	Female	22.1	No	No	Good	Good	No
Mike Johnson	52	Male	28.9	Yes	Yes	Fair	Fair	Yes
Sarah Lee	30	Female	20.3	No	No	Good	Good	No
David Brown	48	Male	26.7	Yes	No	Fair	Fair	No

Step 4: Data Preparation

Procedure

The imported data was verified before visualization.

The following checks were performed:

- Checked Null Values
- Verified Data Types
- Renamed Columns if required
- Classified Dimensions
- Classified Measures

Dimensions

- Age Category
- Gender
- Smoking
- Diabetes
- General Health

Measures

- BMI
- Physical Health
- Mental Health

- Sleep Time

Output

Clean dataset prepared.

Step 5: Create Worksheet 1

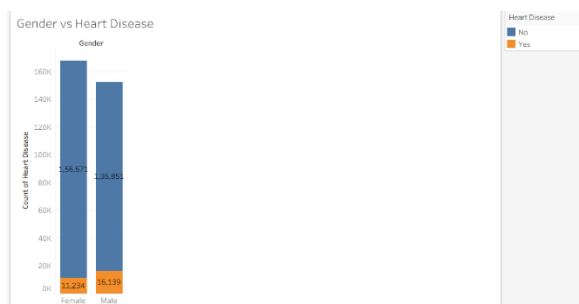
Gender vs Heart Disease

Procedure

1. Open Worksheet.
2. Drag Heart Disease into Columns.
3. Drag Number of Records into Rows.
4. Create a Bar Chart.
5. Add Labels.

Output

Gender vs Heart Disease distribution chart created.



Step 6: Create Worksheet 2

Age vs Heart Disease

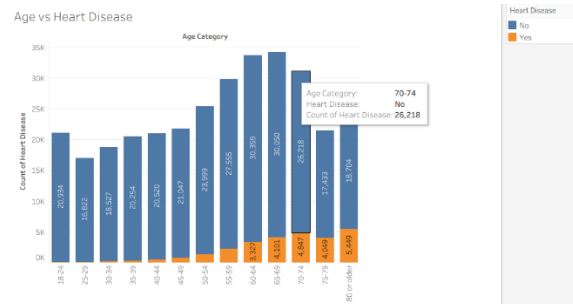
Procedure

1. Drag Age Category to Columns.
2. Drag Number of Records to Rows.
3. Filter using Heart Disease.

4. Apply colors.

Output

Age-wise heart disease analysis completed.



Step 7: Create Worksheet 3

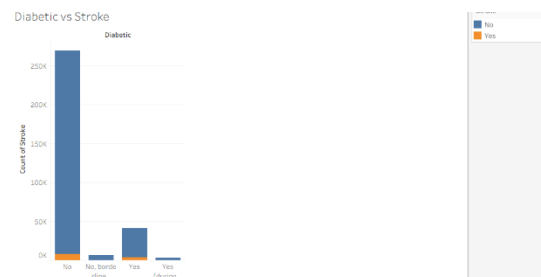
Diabetic vs Stroke

Procedure

1. Drag Sex into Columns.
2. Drag Number of Records into Rows.
3. Color by Heart Disease.
4. Add Labels.

Output

Diabetic vs Stroke chart created.



Step 8: Create Worksheet 4

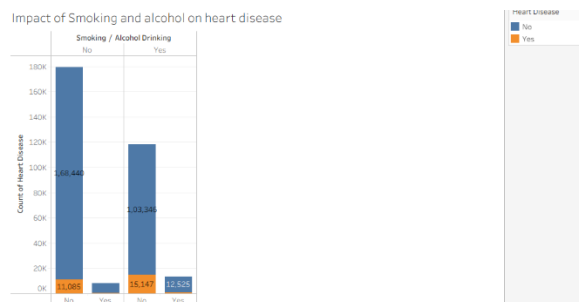
Impact of smoking and alcohol on heart disease

Procedure

1. Drag Smoking into Columns.
2. Drag Heart Disease Count into Rows.
3. Apply Filters.
4. Format the chart.

Output

Smoking and Alcohol impact analyzed.



Step 9: Create Worksheet 5

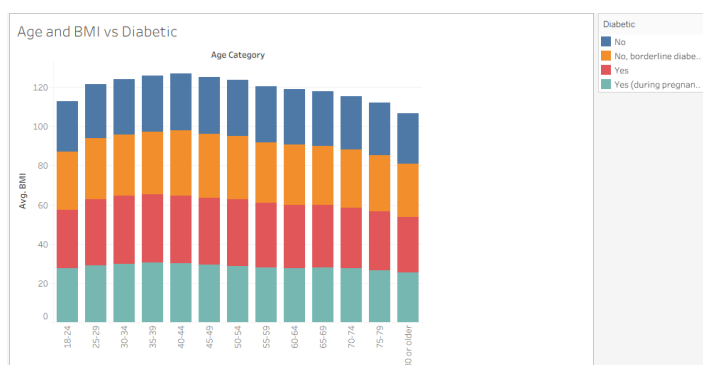
BMI Analysis

Procedure

1. Drag BMI.
2. Create Histogram.
3. Compare with Heart Disease.

Output

BMI distribution created.



Step 10: Create Worksheet 6

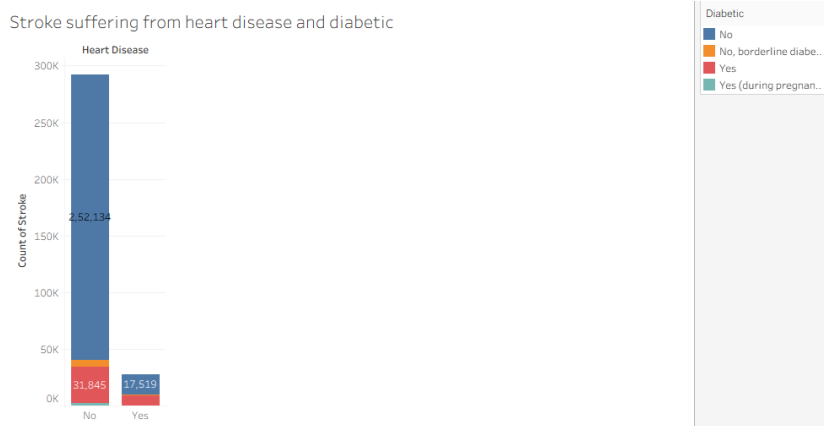
Diabetes Analysis

Procedure

1. Drag Diabetes.
2. Compare with Heart Disease.
3. Use Bar Chart.

Output

Diabetes relationship identified.



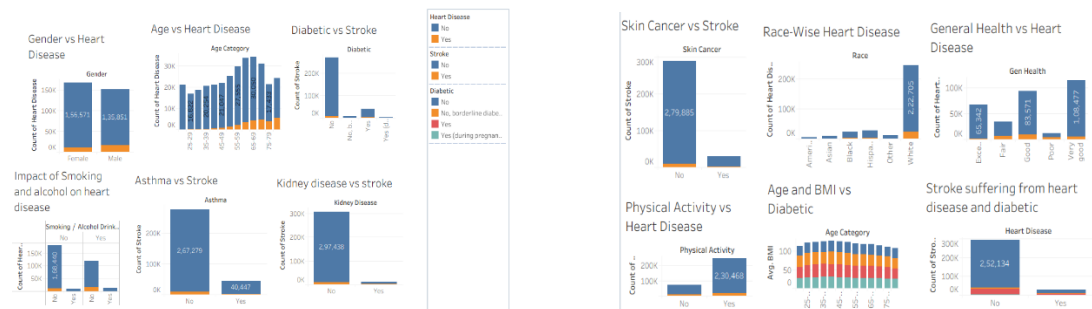
Step 11: Create Dashboard

Procedure

1. Click New Dashboard.
2. Set Dashboard Size.
3. Drag all Worksheets into Dashboard.
4. Arrange Charts.
5. Add Dashboard Title.
6. Add Filters.
7. Enable Interactive Actions.

Output

Interactive dashboard completed.



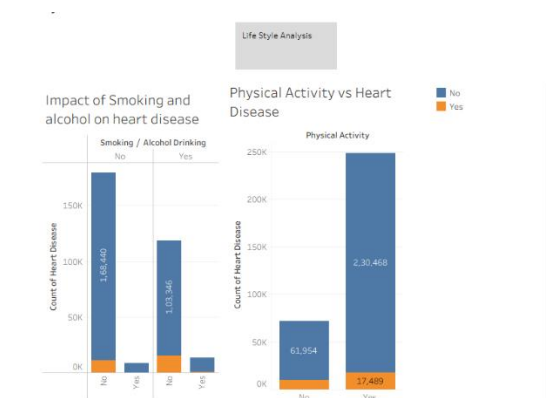
Step 12: Create Story

Procedure

1. Click New Story.
2. Add Dashboard.
3. Create Story Points.
4. Add Captions.
5. Explain Findings.

Output

Story successfully created.



Step 13: Dashboard Testing

Procedure

The dashboard was tested using:

- Filter Selection
- Highlight Actions
- Navigation
- Dynamic Updates

The dashboard responded correctly to all interactions.

Step 14: Results

The analysis identified several key findings:

- Older age groups have higher heart disease prevalence.
- Smokers are more likely to have heart disease.
- Diabetes significantly increases heart disease risk.
- Poor general health is associated with a higher incidence of heart disease.
- Physical activity is linked to a lower risk of heart disease.

Step 15: Conclusion

The Heart Disease Analysis Dashboard was successfully developed using Tableau. The dashboard transforms raw healthcare data into meaningful visual insights, allowing users to explore patterns, compare health indicators, and support informed healthcare decisions through interactive visualizations.